

HYDRA Documentation

Installation Guide Oracle 19c for Windows for HYDRA MW4.0pe

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1 Introduction

Attention:

To perform the following procedures advanced knowledge of your server operating system, the intended database software and common IT systems is required.

These procedures are performed on your own responsibility.

MPDV Mikrolab GmbH is not liable for any loss of or destruction of data.

In case of doubt place an order with MPDV Mikrolab GmbH to perform these tasks for you.

This manual explains how to install Oracle Database 19c (19.3.0.0.0) on a server with a Windows Server 2019 operating system for use with MPDVs Manufacturing Integration Platform (**MIP**) based products like MIP 1.1, HYDRA MW4.0pe or FEDRA 1.1.

Other than MIP the applications HYDRA and FEDRA allow for a multi system installation (multiple HYDRA or FEDRA systems on the same (application) server).

For multi system installations every HYDRA or FEDRA system needs its own database.

Name your databases **mip1**, **mip2**, **mip3**, etc.

With multi system installations it is strongly recommended that every database has its own database instance available, e.g.: **MIP1**, **MIP2**, **MIP3**, etc.

If the database installation is meant as preliminary installation for MPDV personnel to install a MIP based product on your server you can stop as soon as you are finished with chapter "[5 Installing Oracle 64 Bit Client Software](#)".

Attention: Do not run these installation procedures on a computer where MIP based products like MIP 1.1, HYDRA MW4.0pe or FEDRA 1.1 are already running.

By doing so, you might destroy your existing MIP based system.

Please be extremely cautious if you decide to do it anyway.

2 Installation Requirements

- Server hardware and software according to the recommendations of MPDV Mikrolab GmbH for the respective MIP based product.

See manuals: **HW_SW_GUIDE.pdf**

- English operating system according to the recommendations of MPDV Mikrolab GmbH.
- **Important:** Server configuration according to MPDVs server configuration manuals for the respective MIP based product, e.g.:
"Preparation Guide for Windows Server 2019" (e.g.: PreInstGuide_MW40_WIN2019.pdf)

- Installation file for the required edition of **"Oracle Database 19c (19.3) for Microsoft Windows x64 (64-bit)"**

WINDOWS.X64_193000_db_home.zip

e.g.: <https://www.oracle.com/database/technologies/oracle19c-windows-downloads.html#license-lightbox>

Oracle Database 19c (19.3) for Microsoft Windows x64 (64-bit)

Download	Description
 WINDOWS.X64_193000_db_home.zip	(3,105,763,999 bytes) (sha256sum - 64d92018207829833bd4d00f1a7fb40c531c8a4a68ded9e430a5d6fbaedaca95)

- Installation files for the 64-bit version of the **"Oracle Database 19c Client (19.3) for Microsoft Windows x64 (64-bit)"**

WINDOWS.X64_193000_client_home.zip

Download	Description
 WINDOWS.X64_193000_client_home.zip	(64-bit) (924,022,453 bytes) (sha256sum - a7d5ba7d3dac84fceb5c5ddb50aedd3125b52d94f26bab9924bdab6f6d00d2c7e)

- Minimum hard disk configuration:
Partition **C:** with a minimum of **100 GB** disk space available and file system **NTFS** for the Windows installation
Partition **D:** with a minimum of **200 GB** disk space available and file system **NTFS** for the installation of the MIP based product
Partition **E:** with a minimum of **400 GB** disk space available and file system **NTFS** for database data (default configuration)
- Note: If you are using a different disk layout you must make sure that it matches the configuration used in the database installation scripts (see chapter [6.2 Create MIP Database](#)).
- Screen resolution with min. **1024x768** pixel.
- Local Windows user **mipadm**
(see server configuration manual, e.g. PreInstGuide_MW40_WIN2019.pdf)
The user **mipadm** must be member of the local group **Administrators**.

3 Installing Oracle Server Software

Before starting the installation of the Oracle database software it is recommended to review the standard installation documentation from Oracle. Especially regarding the topic installation of multiple Oracle database and client versions of the same hardware.

<https://docs.oracle.com/en/database/oracle/oracle-database/19/ntdbi/database-installation-guide-microsoft-windows.pdf>

<https://docs.oracle.com/en/database/oracle/oracle-database/19/ntcli/database-client-installation-guide-microsoft-windows.pdf>

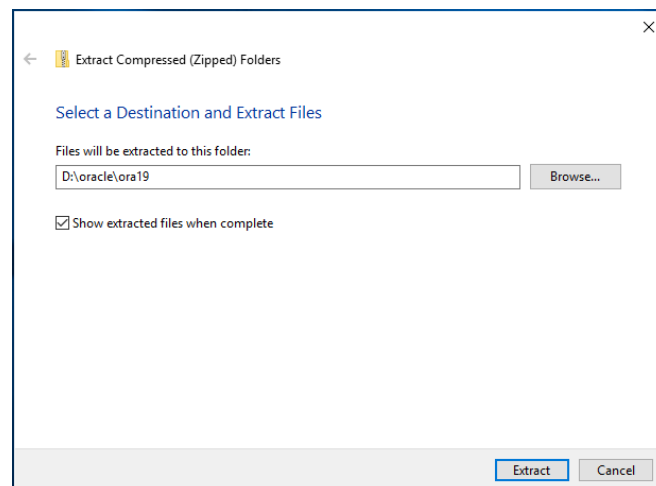
This document only covers the installation of the Oracle 19c database software on a system without any other installed versions.

Login as local user **mipadm**.

Unblock the downloaded zip file (see chapter [2 Installation Requirements](#)) if the download was done with the Internet Explorer.

If the file properties look like below select "**Unblock**" in the file properties of the ZIP file **before extracting** its contents:

Unzip the Oracle installation file to your intended ORACLE_HOME location (e.g.: **D:\oracle\ora19**):



For performance reasons you might want to exclude the destination directory (e.g.: D:\oracle\ora19) from scanning with the Windows Defender and/or other anti-virus software.

After the extraction of the Oracle ZIP file you have to adapt the Oracle database response file before you can proceed with the installation.

The installation will be executed as a silent installation. The response file is stored at

D:\mip1\db_sql\oracle19\windows\db_inst\ora19db_inst.rsp.

Before you could with the installation some (red) values must checked. The preconfigured values comply to the MPDV recommendations. Dissenting installations could be performed in charge of the customer.

```
#####
## Copyright(c) Oracle Corporation 1998,2019. All rights reserved.      ##
##                                                                    ##
## Specify values for the variables listed below to customize           ##
## your installation.                                                    ##
##                                                                    ##
## Each variable is associated with a comment. The comment              ##
## can help to populate the variables with the appropriate              ##
## values.                                                                ##
##                                                                    ##
## IMPORTANT NOTE: This file contains plain text passwords and          ##
## should be secured to have read permission only by oracle user        ##
## or db administrator who owns this installation.                      ##
##                                                                    ##
#####

#-----
# Do not change the following system generated value.
#-----
oracle.install.responseFileVersion=/oracle/install/rspfmt_dbinstall_response_schema_v19.0.0

#-----
# Specify the installation option.
# It can be one of the following:
# - INSTALL_DB_SWONLY
# - INSTALL_DB_AND_CONFIG
#-----
oracle.install.option=INSTALL_DB_SWONLY

#-----
# Specify the Unix group to be set for the inventory directory.
#-----
UNIX_GROUP_NAME=

#-----
# Specify the complete path of the Oracle Base.
#-----
ORACLE_BASE=D:\oracle

#-----
# Specify the installation edition of the component.
#
# The value should contain only one of these choices.

# - EE   : Enterprise Edition

# - SE2   : Standard Edition 2
```

```
#-----
oracle.install.db.InstallEdition=SE2
#####
#
# PRIVILEGED OPERATING SYSTEM GROUPS
# -----
# Provide values for the OS groups to which SYSDBA and SYSOPER privileges
# needs to be granted. If the install is being performed as a member of the
# group "dba", then that will be used unless specified otherwise below.
#
# The value to be specified for OSDBA and OSOPER group is only for UNIX based
# Operating System.
#
#####

#-----
# The OSDBA_GROUP is the OS group which is to be granted SYSDBA privileges.
#-----
oracle.install.db.OSDBA_GROUP=

#-----
# The OSOPER_GROUP is the OS group which is to be granted SYSOPER privileges.
# The value to be specified for OSOPER group is optional.
#-----
oracle.install.db.OSOPER_GROUP=

#-----
# The OSBACKUPDBA_GROUP is the OS group which is to be granted SYSBACKUP privileges.
#-----
oracle.install.db.OSBACKUPDBA_GROUP=

#-----
# The OSDGDBA_GROUP is the OS group which is to be granted SYSDBG privileges.
#-----
oracle.install.db.OSDGDBA_GROUP=

#-----
# The OSKMDBA_GROUP is the OS group which is to be granted SYSKM privileges.
#-----
oracle.install.db.OSKMDBA_GROUP=

#-----
# The OSRACDBA_GROUP is the OS group which is to be granted SYSRAC privileges.
#-----
oracle.install.db.OSRACDBA_GROUP=
#####
#
# Privileged user configuration
#
#####

#####
#
# Grid Options
#
#####

#-----
# Value is required only if the specified install option is INSTALL_DB_SWONLY
#
# Specify the cluster node names selected during the installation.
#
# Example : oracle.install.db.CLUSTER_NODES=node1,node2
#-----
oracle.install.db.CLUSTER_NODES=
```

```
#####
#
#           Database Configuration Options
#
#####

#-----
# Specify the type of database to create.
# It can be one of the following:
# - GENERAL_PURPOSE
# - DATA_WAREHOUSE
# GENERAL_PURPOSE: A starter database designed for general purpose use or transaction-heavy applications.
# DATA_WAREHOUSE : A starter database optimized for data warehousing applications.
#-----
oracle.install.db.config.starterdb.type=GENERAL_PURPOSE

#-----
# Specify the Starter Database Global Database Name.
#-----
oracle.install.db.config.starterdb.globalDBName=

#-----
# Specify the Starter Database SID.
#-----
oracle.install.db.config.starterdb.SID=

#-----
# Specify whether the database should be configured as a Container database.
# The value can be either "true" or "false". If left blank it will be assumed
# to be "false".
#-----
oracle.install.db.ConfigureAsContainerDB=false

#-----
# Specify the Pluggable Database name for the pluggable database in Container Database.
#-----
oracle.install.db.config.PDBName=

#-----
# Specify the Starter Database character set.
#
# One of the following
# AL32UTF8, WE8ISO8859P15, WE8MSWIN1252, EE8ISO8859P2,
# EE8MSWIN1250, NE8ISO8859P10, NEE8ISO8859P4, BLT8MSWIN1257,
# BLT8ISO8859P13, CL8ISO8859P5, CL8MSWIN1251, AR8ISO8859P6,
# AR8MSWIN1256, EL8ISO8859P7, EL8MSWIN1253, IW8ISO8859P8,
# IW8MSWIN1255, JA16EUC, JA16EUCTILDE, JA16SJIS, JA16SJISTILDE,
# KO16MSWIN949, ZHS16GBK, TH8TISASCII, ZHT32EUC, ZHT16MSWIN950,
# ZHT16HKSCS, WE8ISO8859P9, TR8MSWIN1254, VN8MSWIN1258
#-----
oracle.install.db.config.starterdb.characterSet=

#-----
# This variable should be set to true if Automatic Memory Management
# in Database is desired.
# If Automatic Memory Management is not desired, and memory allocation
# is to be done manually, then set it to false.
#-----
oracle.install.db.config.starterdb.memoryOption=false

#-----
# Specify the total memory allocation for the database. Value(in MB) should be
# at least 256 MB, and should not exceed the total physical memory available
# on the system.
# Example: oracle.install.db.config.starterdb.memoryLimit=512
#-----
```

```

oracle.install.db.config.starterdb.memoryLimit=

#-----
# This variable controls whether to load Example Schemas onto
# the starter database or not.
# The value can be either "true" or "false". If left blank it will be assumed
# to be "false".
#-----
oracle.install.db.config.starterdb.installExampleSchemas=false

#####
#
# Passwords can be supplied for the following four schemas in the
# starter database:
# SYS
# SYSTEM
# DBSNMP (used by Enterprise Manager)
#
# Same password can be used for all accounts (not recommended)
# or different passwords for each account can be provided (recommended)
#
#####

#-----
# This variable holds the password that is to be used for all schemas in the
# starter database.
#-----
oracle.install.db.config.starterdb.password.ALL=

#-----
# Specify the SYS password for the starter database.
#-----
oracle.install.db.config.starterdb.password.SYS=

#-----
# Specify the SYSTEM password for the starter database.
#-----
oracle.install.db.config.starterdb.password.SYSTEM=

#-----
# Specify the DBSNMP password for the starter database.
# Applicable only when oracle.install.db.config.starterdb.managementOption=CLOUD_CONTROL
#-----
oracle.install.db.config.starterdb.password.DBSNMP=

#-----
# Specify the PDBADMIN password required for creation of Pluggable Database in the Container Database.
#-----
oracle.install.db.config.starterdb.password.PDBADMIN=

#-----
# Specify the management option to use for managing the database.
# Options are:
# 1. CLOUD_CONTROL - If you want to manage your database with Enterprise Manager Cloud Control along with Database
# Express.
# 2. DEFAULT -If you want to manage your database using the default Database Express option.
#-----
oracle.install.db.config.starterdb.managementOption=DEFAULT

#-----
# Specify the OMS host to connect to Cloud Control.
# Applicable only when oracle.install.db.config.starterdb.managementOption=CLOUD_CONTROL
#-----
oracle.install.db.config.starterdb.omsHost=

#-----
# Specify the OMS port to connect to Cloud Control.

```

```
# Applicable only when oracle.install.db.config.starterdb.managementOption=CLOUD_CONTROL
#-----
oracle.install.db.config.starterdb.omsPort=0

#-----
# Specify the EM Admin user name to use to connect to Cloud Control.
# Applicable only when oracle.install.db.config.starterdb.managementOption=CLOUD_CONTROL
#-----
oracle.install.db.config.starterdb.emAdminUser=

#-----
# Specify the EM Admin password to use to connect to Cloud Control.
# Applicable only when oracle.install.db.config.starterdb.managementOption=CLOUD_CONTROL
#-----
oracle.install.db.config.starterdb.emAdminPassword=

#####
#
# SPECIFY RECOVERY OPTIONS
# -----
# Recovery options for the database can be mentioned using the entries below
#
#####

#-----
# This variable is to be set to false if database recovery is not required. Else
# this can be set to true.
#-----
oracle.install.db.config.starterdb.enableRecovery=false

#-----
# Specify the type of storage to use for the database.
# It can be one of the following:
# - FILE_SYSTEM_STORAGE
# - ASM_STORAGE
#-----
oracle.install.db.config.starterdb.storageType=

#-----
# Specify the database file location which is a directory for datafiles, control
# files, redo logs.
#
# Applicable only when oracle.install.db.config.starterdb.storage=FILE_SYSTEM_STORAGE
#-----
oracle.install.db.config.starterdb.fileSystemStorage.dataLocation=

#-----
# Specify the recovery location.
#
# Applicable only when oracle.install.db.config.starterdb.storage=FILE_SYSTEM_STORAGE
#-----
oracle.install.db.config.starterdb.fileSystemStorage.recoveryLocation=

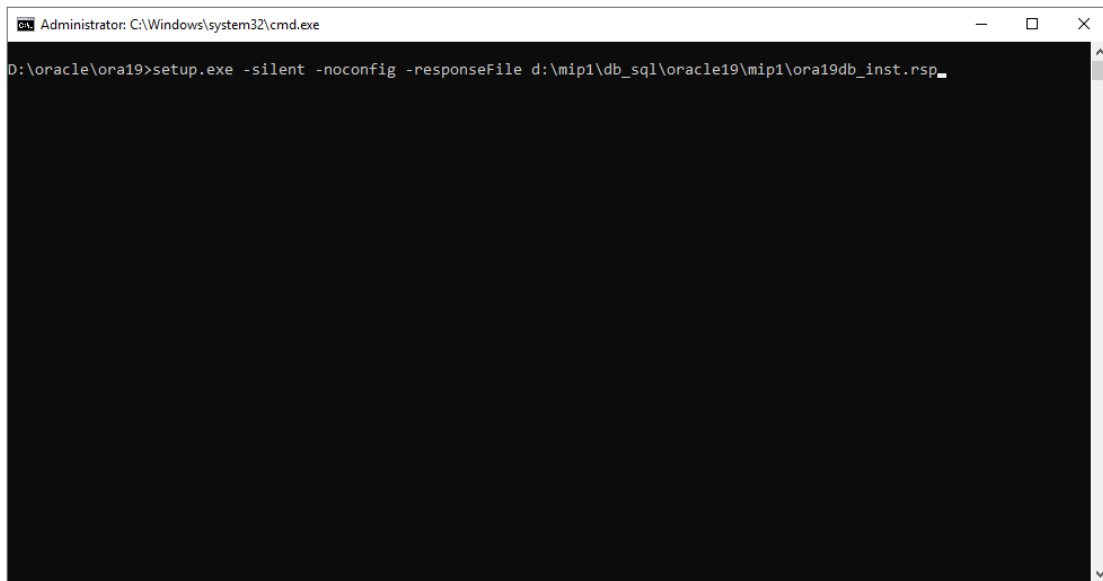
#-----
# Specify the existing ASM disk groups to be used for storage.
#
# Applicable only when oracle.install.db.config.starterdb.storageType=ASM_STORAGE
#-----
oracle.install.db.config.asm.diskGroup=

#-----
# Specify the password for ASMSNMP user of the ASM instance.
#
# Applicable only when oracle.install.db.config.starterdb.storage=ASM_STORAGE
#-----
oracle.install.db.config.asm.ASMNMPPassword=
```

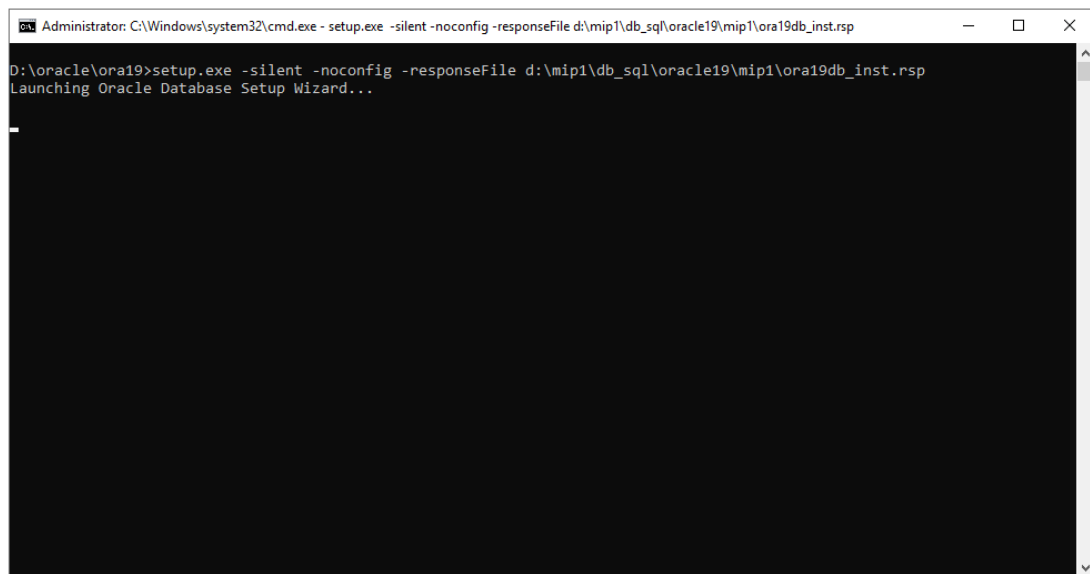
```
#-----  
# Specify the Oracle Home user.  
#  
# Oracle recommends that you specify a Windows User Account with limited privilege to install  
# and configure a secure Oracle home. Set oracle.install.IsVirtualAccount to true  
# if you want to use Virtual Account.  
#  
# Set oracle.install.IsBuiltInAccount and oracle.install.IsVirtualAccount to false if you want to use Windows Account user  
# as Oracle Home user.  
#-----  
oracle.install.IsBuiltInAccount=false  
oracle.install.IsVirtualAccount=false  
oracle.install.OracleHomeUserName=orahome  
oracle.install.OracleHomeUserPassword=ORAHome!!2019
```

Start the installation inside the extracted directory (e.g.: D:\oracle\ora19) in a command line:

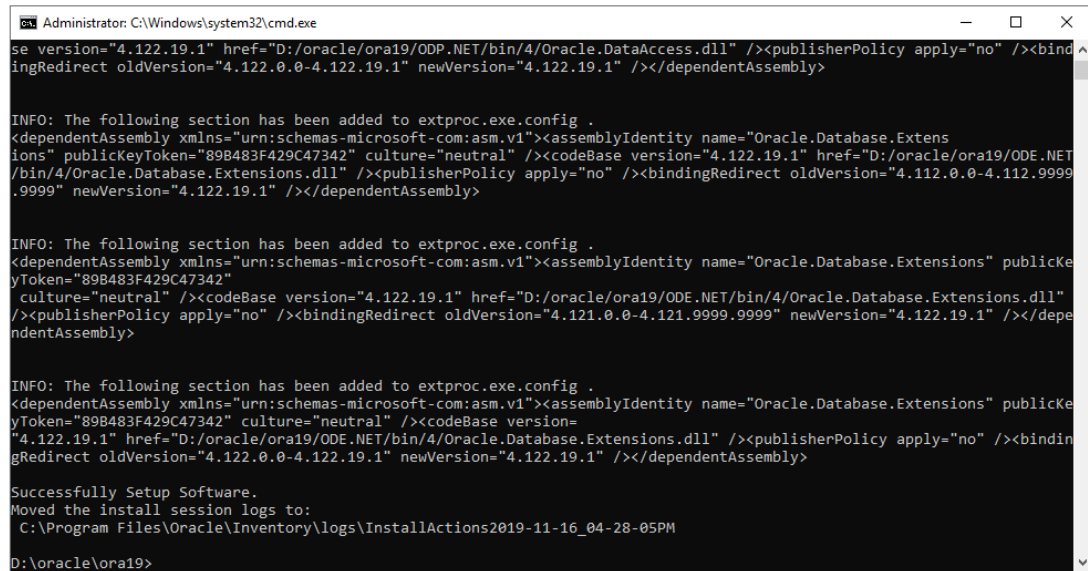
D:\oracle\ora19>**setup.exe -silent -noconfig -responseFile**



```
Administrator: C:\Windows\system32\cmd.exe
D:\oracle\ora19>setup.exe -silent -noconfig -responseFile d:\mip1\db_sql\oracle19\mip1\ora19db_inst.rsp_
```



```
Administrator: C:\Windows\system32\cmd.exe - setup.exe -silent -noconfig -responseFile d:\mip1\db_sql\oracle19\mip1\ora19db_inst.rsp
D:\oracle\ora19>setup.exe -silent -noconfig -responseFile d:\mip1\db_sql\oracle19\mip1\ora19db_inst.rsp
Launching Oracle Database Setup Wizard...
```



```
Administrator: C:\Windows\system32\cmd.exe
se version="4.122.19.1" href="D:/oracle/ora19/ODP.NET/bin/4/Oracle.DataAccess.dll" /><publisherPolicy apply="no" /><bindingRedirect oldVersion="4.122.0.0-4.122.19.1" newVersion="4.122.19.1" /></dependentAssembly>

INFO: The following section has been added to extproc.exe.config .
<dependentAssembly xmlns="urn:schemas-microsoft-com:asm.v1"><assemblyIdentity name="Oracle.Database.Extensions" publicKeyToken="89B483F429C47342" culture="neutral" /><codeBase version="4.122.19.1" href="D:/oracle/ora19/ODP.NET/bin/4/Oracle.Database.Extensions.dll" /><publisherPolicy apply="no" /><bindingRedirect oldVersion="4.122.0.0-4.122.19.1" newVersion="4.122.19.1" /></dependentAssembly>

INFO: The following section has been added to extproc.exe.config .
<dependentAssembly xmlns="urn:schemas-microsoft-com:asm.v1"><assemblyIdentity name="Oracle.Database.Extensions" publicKeyToken="89B483F429C47342" culture="neutral" /><codeBase version="4.122.19.1" href="D:/oracle/ora19/ODP.NET/bin/4/Oracle.Database.Extensions.dll" /><publisherPolicy apply="no" /><bindingRedirect oldVersion="4.122.0.0-4.122.19.1" newVersion="4.122.19.1" /></dependentAssembly>

INFO: The following section has been added to extproc.exe.config .
<dependentAssembly xmlns="urn:schemas-microsoft-com:asm.v1"><assemblyIdentity name="Oracle.Database.Extensions" publicKeyToken="89B483F429C47342" culture="neutral" /><codeBase version="4.122.19.1" href="D:/oracle/ora19/ODP.NET/bin/4/Oracle.Database.Extensions.dll" /><publisherPolicy apply="no" /><bindingRedirect oldVersion="4.122.0.0-4.122.19.1" newVersion="4.122.19.1" /></dependentAssembly>

Successfully Setup Software.
Moved the install session logs to:
C:\Program Files\Oracle\Inventory\logs\InstallActions2019-11-16_04-28-05PM
D:\oracle\ora19>
```

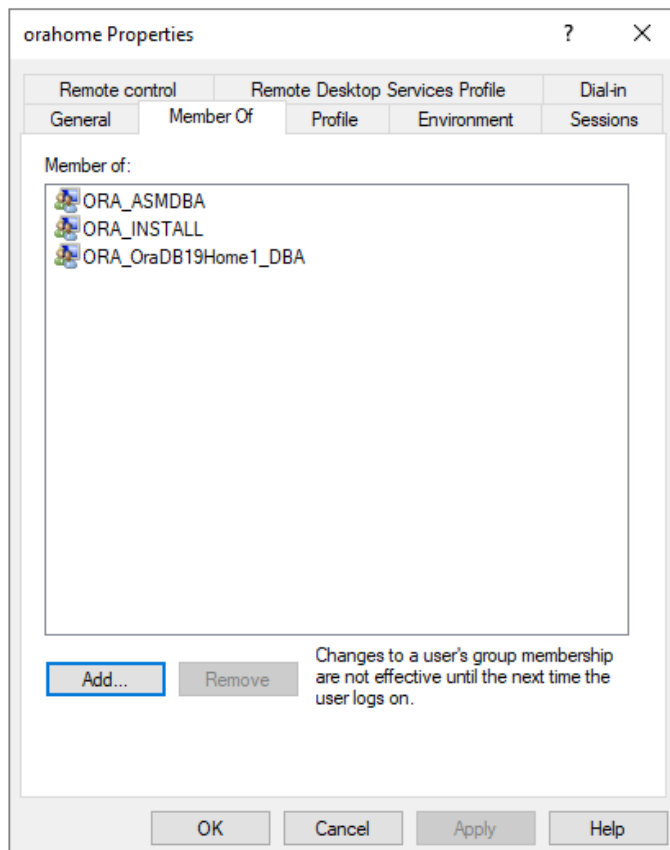
Check the end of command line output: “Successfully Setup Software”

3.1 Users and Groups

The following groups are created when installing Oracle 19c:

 ORA_ASMDBA	ORA_ASMDBA
 ORA_ASMOPER	ORA_ASMOPER
 ORA_CLIENT_LISTENERS	ORA_CLIENT_LISTENERS
 ORA_DBA	ORA_DBA
 ORA_DBSVCACCTS	ORA_DBSVCACCTS
 ORA_GRID_LISTENERS	ORA_GRID_LISTENERS
 ORA_INSTALL	ORA_INSTALL
 ORA_OPER	ORA_OPER
 ORA_OraDB19Home1_DBA	ORA_OraDB19Home1_DBA
 ORA_OraDB19Home1_OPER	ORA_OraDB19Home1_OPER
 ORA_OraDB19Home1_SYSDG	ORA_OraDB19Home1_SYSDG
 ORA_OraDB19Home1_SYSKM	ORA_OraDB19Home1_SYSKM

User orahome was added as additional Windows User:



4 Installing Oracle Patch Set

- The latest Oracle patch set approved by MPDV Mikrolab GmbH must be installed.
Version: **currently there is no approved patch set to be installed**
- Please take note of the "Patch Set Notes" delivered with the patch set.

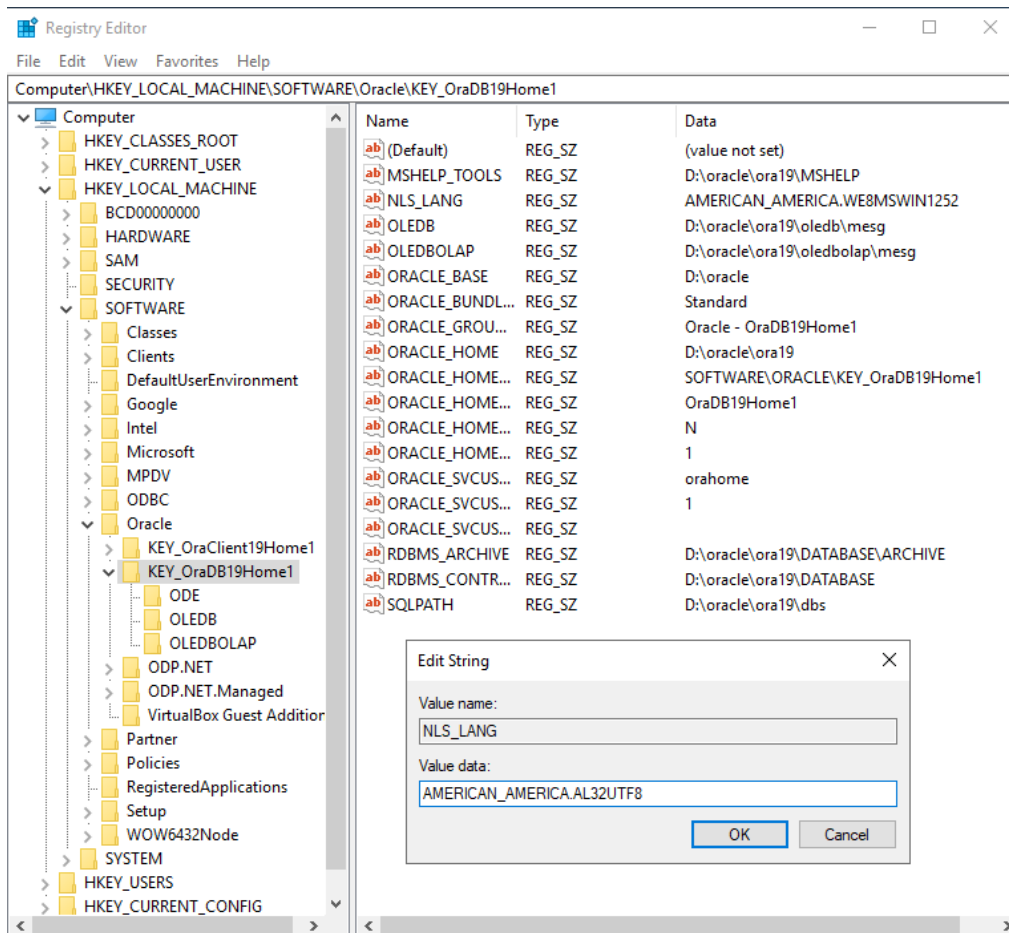
5 Post Installation Configuration

5.1 Registry Settings

Change the Windows Registry (regedit):

in **HKEY_LOCAL_MACHINE\SOFTWARE\ORACLE\KEY_OraDB19Home1** the contents for key **NLS_LANG** must be changed:

new: **AMERICAN_AMERICA.AL32UTF8**



Should **HKEY_LOCAL_MACHINE\SOFTWARE\ORACLE*** contain other entries for **NLS_LANG** change them as well.

5.2 Network Connectivity

To perform the following steps either the MIP based application software must already be available on the server, the following files must be provided to you by MPDV or you need to create those files from scratch!

Starting with HYDRA MW4.0pe every instance will have its own Oracle Listener configuration.

Copy the files from directory `d:\mip1\db_sql\oracle19\windows\network\server\*` to directory `d:\oracle\ora19\network\admin\*`:

Check and edit the configuration file of the Oracle Listener:

d:\oracle\ora19\network\admin\listener.ora

```
# listener.ora Network Configuration File: D:\oracle\ora19\NETWORK\ADMIN\listener.ora
# Generated by Oracle configuration tools.
```

```
SID_LIST_LISTENER_MIP1 =
  (SID_LIST =
    (SID_DESC =
      (SID_NAME = CLRExtProc)
      (ORACLE_HOME = D:\oracle\ora19)
      (PROGRAM = extproc)
      (ENVS = "EXTPROC_DLLS=ONLY:D:\oracle\ora19\bin\oraclr19.dll")
    )
  )

LISTENER_MIP1 =
  (DESCRIPTION_LIST =
    (DESCRIPTION =
      (ADDRESS = (PROTOCOL = TCP) (HOST = [FQDN NAME]) (PORT = 1521))
      (ADDRESS = (PROTOCOL = IPC) (KEY = EXTPROC1521))
    )
  )

LISTENER_MIP2 =
  (DESCRIPTION_LIST =
    (DESCRIPTION =
      (ADDRESS = (PROTOCOL = TCP) (HOST = [FQDN NAME]) (PORT = 1522))
    )
  )
```

Attention!

For [FQDN NAME] you should use the hostname (FQDN) instead of the IP-address.

Check the tnsnames.ora file and add necessary alias configurations for the HYDRA databases (Default: MIP1, MIP2):

d:\oracle\ora19\network\admin\tnsnames.ora

```
# tnsnames.ora Network Configuration File: D:\oracle\ora19\NETWORK\ADMIN\tnsnames.ora
# Generated by Oracle configuration tools.

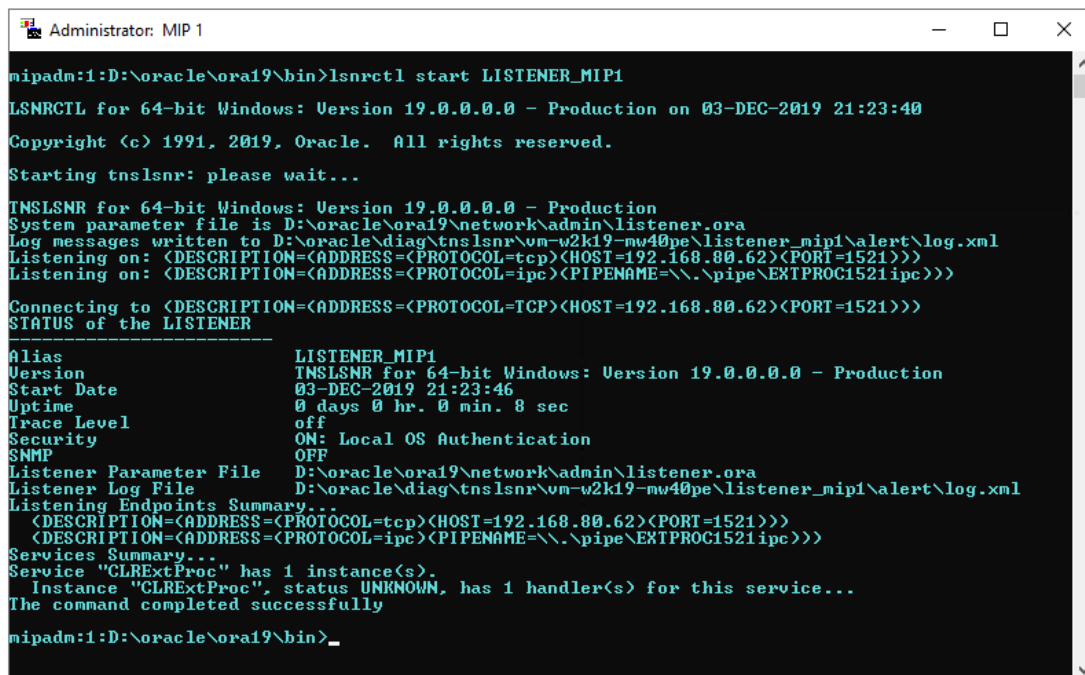
MIP1 =
  (DESCRIPTION =
    (ADDRESS_LIST =
      (ADDRESS = (PROTOCOL = TCP)(HOST = [IP ADDRESS or FQDN NAME])(PORT = 1521))
    )
    (CONNECT_DATA =
      (SERVICE_NAME = mip1)
    )
  )

MIP2 =
  (DESCRIPTION =
    (ADDRESS_LIST =
      (ADDRESS = (PROTOCOL = TCP)(HOST = [IP ADDRESS or FQDN NAME])(PORT = 1522))
    )
    (CONNECT_DATA =
      (SERVICE_NAME = mip2)
    )
  )

ORACLR_CONNECTION_DATA =
  (DESCRIPTION =
    (ADDRESS_LIST =
      (ADDRESS = (PROTOCOL = IPC)(KEY = EXTPROC1521))
    )
    (CONNECT_DATA =
      (SID = CLRExtProc)
      (PRESENTATION = RO)
    )
  )
```

Start a Windows Command Prompt and run the following command:

```
D:\oracle\ora19\bin>lsnrctl start LISTENER_MIP1
```



```
Administrator: MIP 1

mipadm:1:D:\oracle\ora19\bin>lsnrctl start LISTENER_MIP1

LSNRCTL for 64-bit Windows: Version 19.0.0.0.0 - Production on 03-DEC-2019 21:23:40

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Starting tnslnsr: please wait...

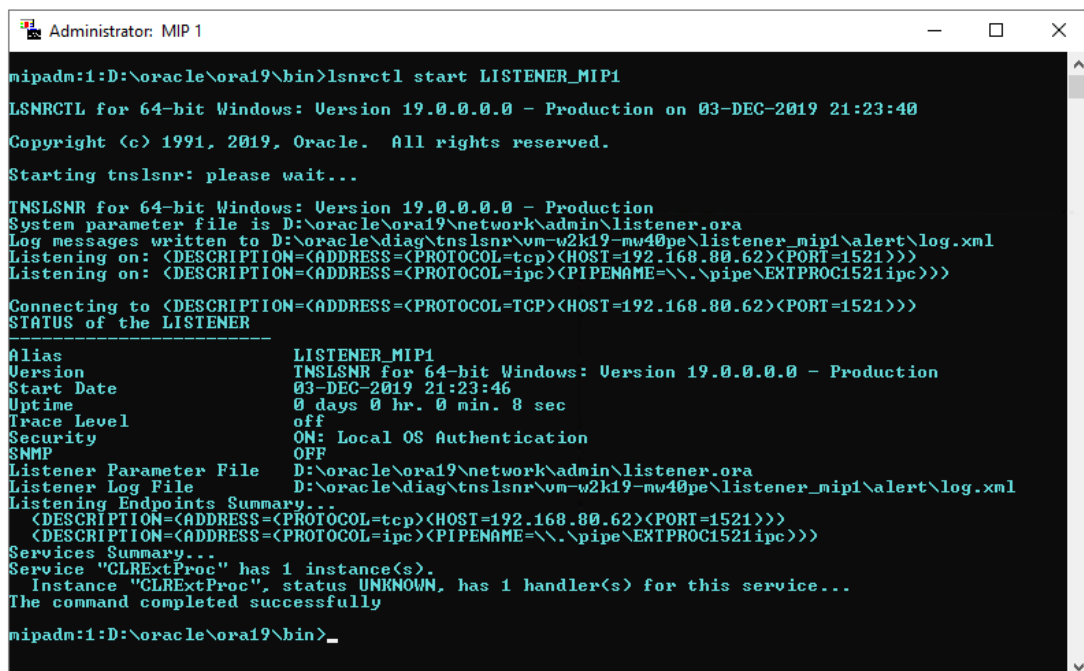
TNSLSNR for 64-bit Windows: Version 19.0.0.0.0 - Production
System parameter file is D:\oracle\ora19\network\admin\listener.ora
Log messages written to D:\oracle\diag\tnslsnr\vm-w2k19-mw40pe\listener_mip1\alert\log.xml
Listening on: (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=192.168.80.62)(PORT=1521)))
Listening on: (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc)(PIPENAME=\\.\pipe\EXTPROC1521ipc)))

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=192.168.80.62)(PORT=1521)))
STATUS of the LISTENER
-----
Alias                     LISTENER_MIP1
Version                  TNSLSNR for 64-bit Windows: Version 19.0.0.0.0 - Production
Start Date               03-DEC-2019 21:23:46
Uptime                   0 days 0 hr. 0 min. 8 sec
Trace Level              off
Security                 ON: Local OS Authentication
SNMP                     OFF
Listener Parameter File  D:\oracle\ora19\network\admin\listener.ora
Listener Log File        D:\oracle\diag\tnslsnr\vm-w2k19-mw40pe\listener_mip1\alert\log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=192.168.80.62)(PORT=1521)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc)(PIPENAME=\\.\pipe\EXTPROC1521ipc)))
Services Summary...
Service "CLRExtProc" has 1 instance(s).
  Instance "CLRExtProc", status UNKNOWN, has 1 handler(s) for this service...
The command completed successfully

mipadm:1:D:\oracle\ora19\bin>_
```

Check the Listener Status:

```
D:\oracle\ora19\bin>lsnrctl status LISTENER_MIP1
```



```
Administrator: MIP 1

mipadm:1:D:\oracle\ora19\bin>lsnrctl start LISTENER_MIP1

LSNRCTL for 64-bit Windows: Version 19.0.0.0.0 - Production on 03-DEC-2019 21:23:40

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Starting tnslnsr: please wait...

TNSLSNR for 64-bit Windows: Version 19.0.0.0.0 - Production
System parameter file is D:\oracle\ora19\network\admin\listener.ora
Log messages written to D:\oracle\diag\tnslsnr\vm-w2k19-mw40pe\listener_mip1\alert\log.xml
Listening on: (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=192.168.80.62)(PORT=1521)))
Listening on: (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc)(PIPENAME=\\.\pipe\EXTPROC1521ipc)))

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=192.168.80.62)(PORT=1521)))
STATUS of the LISTENER
-----
Alias                     LISTENER_MIP1
Version                  TNSLSNR for 64-bit Windows: Version 19.0.0.0.0 - Production
Start Date               03-DEC-2019 21:23:46
Uptime                   0 days 0 hr. 0 min. 8 sec
Trace Level              off
Security                 ON: Local OS Authentication
SNMP                     OFF
Listener Parameter File  D:\oracle\ora19\network\admin\listener.ora
Listener Log File        D:\oracle\diag\tnslsnr\vm-w2k19-mw40pe\listener_mip1\alert\log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=192.168.80.62)(PORT=1521)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc)(PIPENAME=\\.\pipe\EXTPROC1521ipc)))
Services Summary...
Service "CLRExtProc" has 1 instance(s).
  Instance "CLRExtProc", status UNKNOWN, has 1 handler(s) for this service...
The command completed successfully

mipadm:1:D:\oracle\ora19\bin>_
```

6 MIP Database

Every MIP system needs its own MIP database.

Every MIP database needs its own database instance.

Name your MIP databases “mip1”, “mip2”, “mip3”, etc.

Name your MIP database instances “MIP1”, “MIP2”, “MIP3”, etc.

Do not proceed with this chapter if the database installation was meant as preliminary installation for MPDV personnel to install a MIP based product on your server.

Before you might proceed with creating the MIP database(s) on your own you need to have four files available for each MIP database and database instance you want to create.

Example files for MIP database and database instance 1:

CreateMIP1_ora19_win.sql

MIP1_ora19_win.bat

MIP1_ora19_win.sql

initMIP1.ora

Either those files were provided to you in advance by MPDV or you might have the HYDRA application installed at least as far as chapter “4.5 MIP Database” according to the HYDRA installation manual:

InstallationGuide_MW40_Windows.pdf

Note:

The installation manual for HYDRA MW4.0pe will be provided to selected customers only.

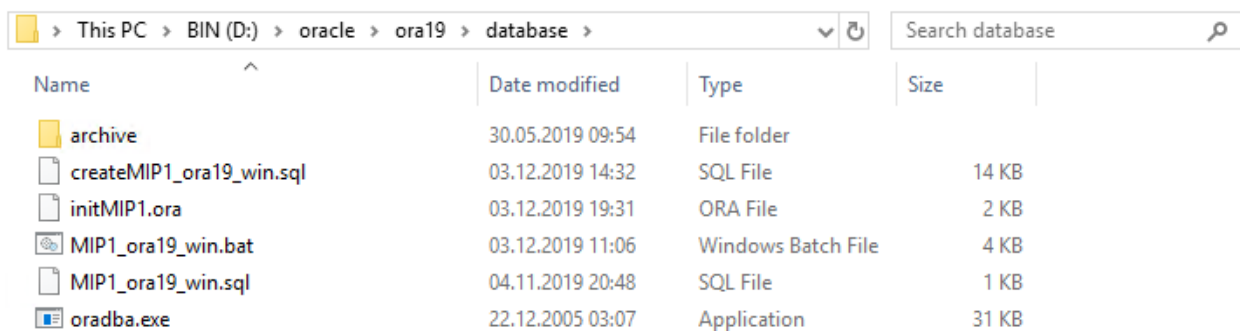
6.1 Create MIP Database

Attention:

Do not run these installation procedures on a computer where a MIP based product is already running. By doing so, you might destroy your existing MIP based system. Please be extremely cautious if you decide to do it anyway.

Logon to the server as local user “mipadm”.

Copy the following four files from `d:\mip1\db_sql\oracle19\windows\mip1\*` to directory `%ORACLE_HOME%\database`, e.g. `d:\oracle\ora19\database\`.



Name	Date modified	Type	Size
archive	30.05.2019 09:54	File folder	
createMIP1_ora19_win.sql	03.12.2019 14:32	SQL File	14 KB
initMIP1.ora	03.12.2019 19:31	ORA File	2 KB
MIP1_ora19_win.bat	03.12.2019 11:06	Windows Batch File	4 KB
MIP1_ora19_win.sql	04.11.2019 20:48	SQL File	1 KB
oradba.exe	22.12.2005 03:07	Application	31 KB

Edit the scripts and make sure that all path configurations are set corresponding to your system environment.

All paths used in the database creation scripts must exist on your server. They will be interactive created by the script `MIP1_ora19_win.bat`.

Do not change or remove settings regarding user configurations and authorizations or any other permissions without consulting with MPDV first!

Note about database user name “mipadm”:

MPDV strongly recommends not to change MIPs default database user name!

Changing the default database user name is not supported when using Oracle databases!

With Oracle databases it is **not** possible to use different database user names other than MIPs default (CREATE USER MIPADM).

Note about password for database user “mipadm”:

For security reasons MPDV strongly recommends to use a secure password for the database user mipadm!

The new password must be disclosed to MPDV.

The following characters are **not** allowed for the password:

- Characters with ASCII Code > 126 (e.g. German umlauts, French “umlauts”, etc.)
- Pipe: |
- Ampersand: &

Inside initMIP1x.ora check the memory settings “sga_target”/“pga_aggregate_target” and adjust it if necessary.

Depending on the total amount of RAM memory available on the server and the system requirements by the MIP based product please set the size for “sga_target” so that as much memory as possible will be used by the database instance

The size of “sga_target” will have direct influence on the performance of the MIP based application because with bigger sizes more data can be cached by the database.

As long as it is ensured that there will be enough RAM left for the requirements of the server’s operating system (~2GB), the MIP based application services (depending on the system size ~1,5-8GB per instance), the MIP WSP service (depending on the system size ~2-12GB per instance) and other database instances installed on the same server you should use as much memory for “sga_target” as possible.

Make sure that the available server memory (RAM) is used to full capacity without forcing your server into swapping.

To install additional database instances, e.g. MIP2, there are corresponding scripts available (d:\mip1\db_sql\oracle19\windows\mip2)

When installing additional database instances make sure to change the port settings for “EM Express” in file “createMIPx_ora19_win.sql” so that there will be no port conflicts, e.g.:

MIP1: `exec dbms_xdb_config.sethttpsport(5500);`

MIP2: `exec dbms_xdb_config.sethttpsport(5501);`

```
-- create EM Express
-----
connect "SYS"/"&&sysPassword" as SYSDBA
spool D:\oracle\admin\mip1\scripts\l2-em_express.log
exec dbms_xdb_config.sethttpsport(5500);
shutdown immediate;
connect "SYS"/"&&sysPassword" as SYSDBA
startup ;
spool off
exit;
```

Open a Windows Command Prompt (cmd.exe).

Change to directory %ORACLE_HOME%\database, e.g.:

```
cd d:\oracle\ora19\database
```

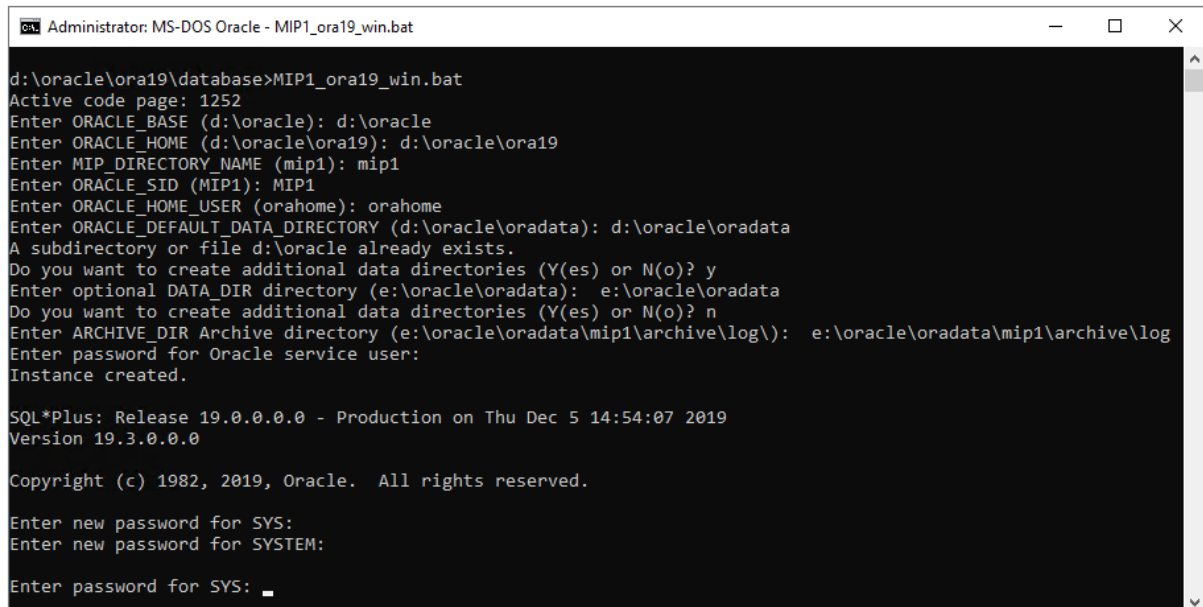
Execute file MIP1_ora19_win.bat and enter the following information. The values in parentheses are default values and have to be insert:

```
Enter ORACLE_BASE (d:\oracle): d:\oracle
Enter ORACLE_HOME (d:\oracle\ora19): d:\oracle\ora19
Enter MIP_DIRECTORY_NAME (mip1): mip1
Enter ORACLE_SID (MIP1): MIP1
Enter ORACLE_HOME_USER (orahome): orahome
Enter ORACLE_DEFAULT_DATA_DIRECTORY (d:\oracle\oradata): d:\oracle\oradata
Do you want to create additional data directories (Y(es) or N(o)? y
Enter optional DATA_DIR directory (e:\oracle\oradata): e:\oracle\oradata
Do you want to create additional data directories (Y(es) or N(o)? n
Enter ARCHIVE_DIR Archive directory (e:\oracle\oradata\mip1\archive\log\):
e:\oracle\oradata\mip1\archive\log

Enter password for Oracle service user: ORAhome!!2019 (see above)
Enter new password for SYS: MW40pe!!2019
Enter new password for SYSTEM: MW40pe!!2019
Enter password for SYS: MW40pe!!2019
```

For security reasons you should want to use different passwords for user SYS and SYSTEM.
Please make sure to disclose the new passwords to MPDV.

Example for installation of database "MIP1"



```
Administrator: MS-DOS Oracle - MIP1_ora19_win.bat
d:\oracle\ora19\database>MIP1_ora19_win.bat
Active code page: 1252
Enter ORACLE_BASE (d:\oracle): d:\oracle
Enter ORACLE_HOME (d:\oracle\ora19): d:\oracle\ora19
Enter MIP_DIRECTORY_NAME (mip1): mip1
Enter ORACLE_SID (MIP1): MIP1
Enter ORACLE_HOME_USER (orahome): orahome
Enter ORACLE_DEFAULT_DATA_DIRECTORY (d:\oracle\oradata): d:\oracle\oradata
A subdirectory or file d:\oracle already exists.
Do you want to create additional data directories (Y(es) or N(o))? y
Enter optional DATA_DIR directory (e:\oracle\oradata): e:\oracle\oradata
Do you want to create additional data directories (Y(es) or N(o))? n
Enter ARCHIVE_DIR Archive directory (e:\oracle\oradata\mip1\archive\log): e:\oracle\oradata\mip1\archive\log
Enter password for Oracle service user:
Instance created.

SQL*Plus: Release 19.0.0.0.0 - Production on Thu Dec 5 14:54:07 2019
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

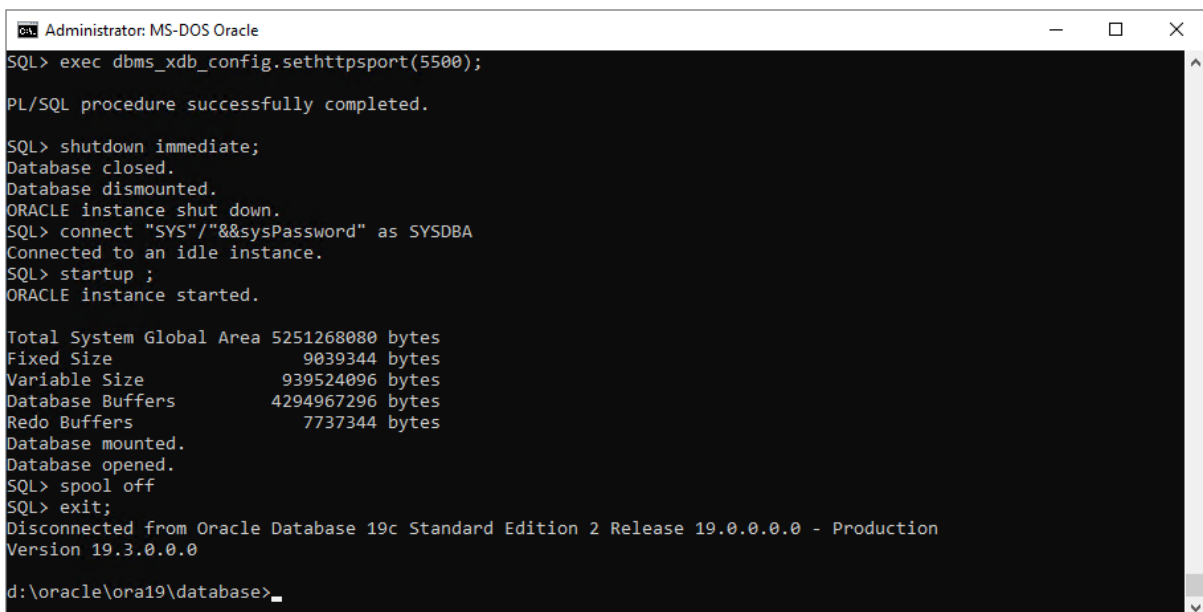
Enter new password for SYS:
Enter new password for SYSTEM:

Enter password for SYS: _
```

Depending on your server environment the installation will run for at least 20 – 30 minutes or longer.

Under no circumstances should you interrupt the installation process!

The installation is finished as soon as "create EM Express" is through and the Windows prompt is visible again, e.g.:



```
Administrator: MS-DOS Oracle
SQL> exec dbms_xdb_config.sethttpsport(5500);

PL/SQL procedure successfully completed.

SQL> shutdown immediate;
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL> connect "SYS"/"&&sysPassword" as SYSDBA
Connected to an idle instance.
SQL> startup ;
ORACLE instance started.

Total System Global Area 5251268080 bytes
Fixed Size 9039344 bytes
Variable Size 939524096 bytes
Database Buffers 4294967296 bytes
Redo Buffers 7737344 bytes
Database mounted.
Database opened.
SQL> spool off
SQL> exit;
Disconnected from Oracle Database 19c Standard Edition 2 Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

d:\oracle\ora19\database>_
```

The database creation must complete without errors.

Check all log files of the installation which can be found in the following directory:

%ORACLE_BASE%\admin\<database name>\scripts

Make a backup copy of all the files you used to create your HYDRA database, e.g.:

```
copy d:\oracle\ora19\database\createMIP1_ora19_win.sql d:\mip1\db_sql
copy d:\oracle\ora19\database\MIP1_ora19_win.bat d:\mip1\db_sql
copy d:\oracle\ora19\database\MIP1_ora19_win.sql d:\mip1\db_sql
copy d:\oracle\ora19\database\initMIP1.ora d:\mip1\db_sql
```

Repeat these procedures for every HYDRA database and its instance.

Should there have been a problem while creating the database you might need to run the whole process again after you fixed the problems which occurred the first time.

It might be that the database is already running after a faulty attempt.

Before you can try it a second time you need to shut down the database first and you need to delete an already created database instance and already created control files.

Shut down the database within a Windows Command Prompt:

```
d:\oracle\ora19\bin\sqlplus /nolog
connect /as sysdba
SQL> shutdown abort
SQL> exit
```

Delete Oracle instance MIP1 within a Windows Command Prompt:

```
d:\oracle\ora19\bin\oradim -delete -sid MIP1
```

Delete already created control files, e.g.:

```
del d:\oracle\oradata\mip1\MIP1_control01.ctl
del d:\oracle\fast_recovery_area\mip1\MIP1_control02.ctl
```

Check if the relevant Windows services were deleted.

 OracleServiceMIP1	Running	Automatic	.\orahome
 OracleVssWriterMIP1	Running	Automatic	.\orahome

Now you might run the database creation process again.

Check the Windows service for the Oracle services

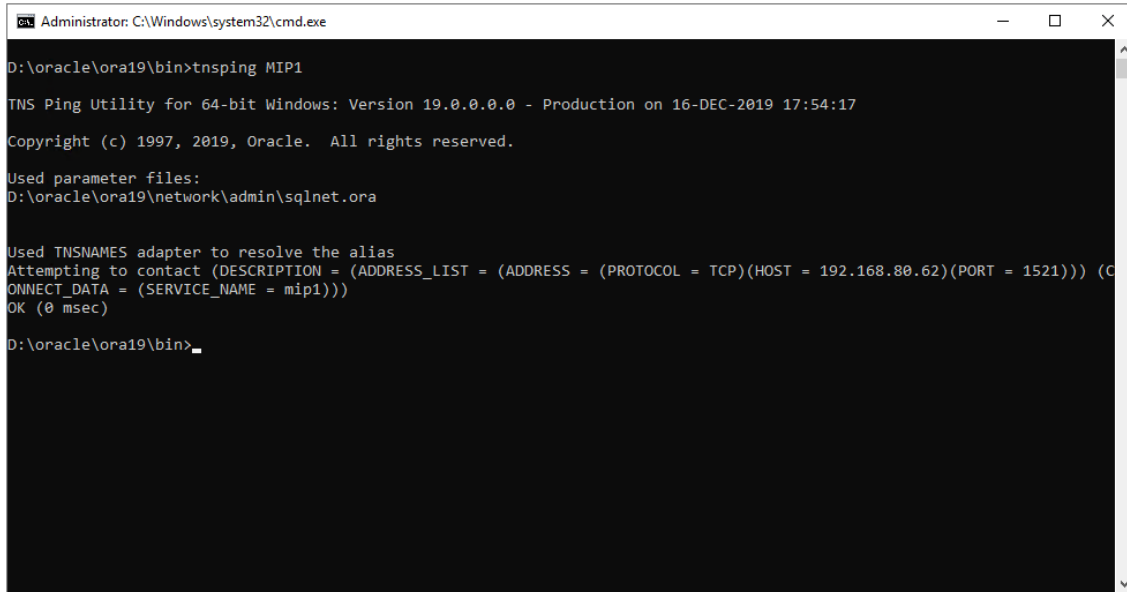
 OracleJobSchedulerMIP1	Disabled		.\orahome
 OracleJobSchedulerMIP2	Disabled		.\orahome
 OracleOraDB19Home1TNSListenerLISTENER_MIP1	Running	Automatic	.\orahome
 OracleOraDB19Home1TNSListenerLISTENER_MIP2	Running	Automatic	.\orahome
 OracleServiceMIP1	Running	Automatic	.\orahome
 OracleServiceMIP2	Running	Automatic	.\orahome
 OracleVssWriterMIP1	Running	Automatic	.\orahome
 OracleVssWriterMIP2	Running	Automatic	.\orahome

6.2 Check Database Connection

Check that the new database is accessible using the **tnsping** command.

Open a Windows command prompt:

```
d:\oracle\ora19\bin>tnsping mip1
```



```
Administrator: C:\Windows\system32\cmd.exe
D:\oracle\ora19\bin>tnsping MIP1

TNS Ping Utility for 64-bit Windows: Version 19.0.0.0.0 - Production on 16-DEC-2019 17:54:17

Copyright (c) 1997, 2019, Oracle. All rights reserved.

Used parameter files:
D:\oracle\ora19\network\admin\sqlnet.ora

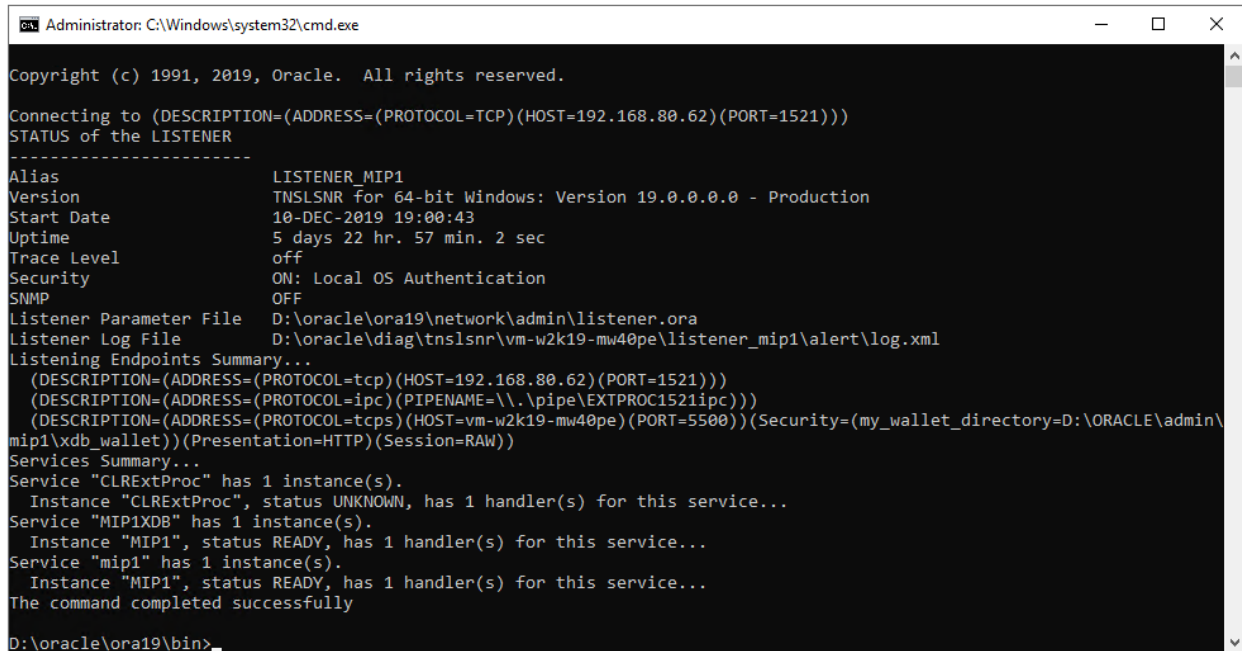
Used TNSNAMES adapter to resolve the alias
Attempting to contact (DESCRIPTION = (ADDRESS_LIST = (ADDRESS = (PROTOCOL = TCP)(HOST = 192.168.80.62)(PORT = 1521))) (C
ONNECT_DATA = (SERVICE_NAME = mip1)))
OK (0 msec)

D:\oracle\ora19\bin>
```

6.3 Check Oracle Listener

After installing the HYDRA database(s) MIP1 (MIP2, MIP3, etc.) check that all services are running using the **lsnrctl** command.

```
d:\oracle\ora19\bin\lsnrctl status LISTENER_MIP1
```



```
Administrator: C:\Windows\system32\cmd.exe

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=192.168.80.62)(PORT=1521)))
STATUS of the LISTENER
-----
Alias                     LISTENER_MIP1
Version                   TNSLSNR for 64-bit Windows: Version 19.0.0.0 - Production
Start Date                10-DEC-2019 19:00:43
Uptime                    5 days 22 hr. 57 min. 2 sec
Trace Level               off
Security                  ON: Local OS Authentication
SNMP                      OFF
Listener Parameter File   D:\oracle\ora19\network\admin\listener.ora
Listener Log File         D:\oracle\diag\tnslsnr\vm-w2k19-mw40pe\listener_mip1\alert\log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=192.168.80.62)(PORT=1521)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc)(PIPENAME=\\.\pipe\EXTPROC1521ipc)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcps)(HOST=vm-w2k19-mw40pe)(PORT=5500))(Security=(my_wallet_directory=D:\ORACLE\admin\
mip1\xdb_wallet))(Presentation=HTTP)(Session=RAW))
Services Summary...
Service "CLRExtProc" has 1 instance(s).
  Instance "CLRExtProc", status UNKNOWN, has 1 handler(s) for this service...
Service "MIP1XDB" has 1 instance(s).
  Instance "MIP1", status READY, has 1 handler(s) for this service...
Service "mip1" has 1 instance(s).
  Instance "MIP1", status READY, has 1 handler(s) for this service...
The command completed successfully

D:\oracle\ora19\bin>
```

Check if a service “mip1” is available

Additional listener could be called by its service name (LISTENER_MIP2, LISTENER_MIP2)

6.4 Check Automatic Startup Database

When you start your server the database(s) must start automatically.

Make sure that “Startup Type” is set to “**Automatic**” for at least the following services:

Database(s): OracleServiceMIP^x

Oracle Listener: OracleOraDB19Home1TNSListenerLISTENER_MIP^x

Name	Description	Status	Startup Type	Log On As
 OracleJobSchedulerMIP1			Disabled	.\orahome
 OracleJobSchedulerMIP2			Disabled	.\orahome
 OracleOraDB19Home1TNSListenerLISTENER_MIP1		Running	Automatic	.\orahome
 OracleOraDB19Home1TNSListenerLISTENER_MIP2		Running	Automatic	.\orahome
 OracleOraDB19Home1TNSListenerLISTENER_MIP3		Running	Automatic	.\orahome
 OracleServiceMIP1		Running	Automatic	.\orahome
 OracleServiceMIP2		Running	Automatic	.\orahome
 OracleVssWriterMIP1		Running	Automatic	.\orahome
 OracleVssWriterMIP2		Running	Automatic	.\orahome

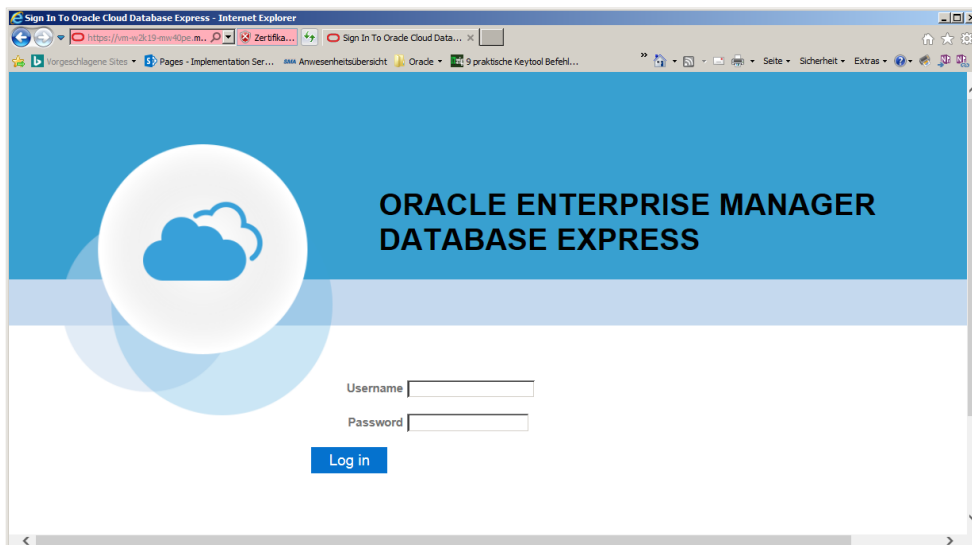
As soon as you are finished with the installation check that the automatic startup is working correctly.

6.5 Check Oracle Enterprise Manager

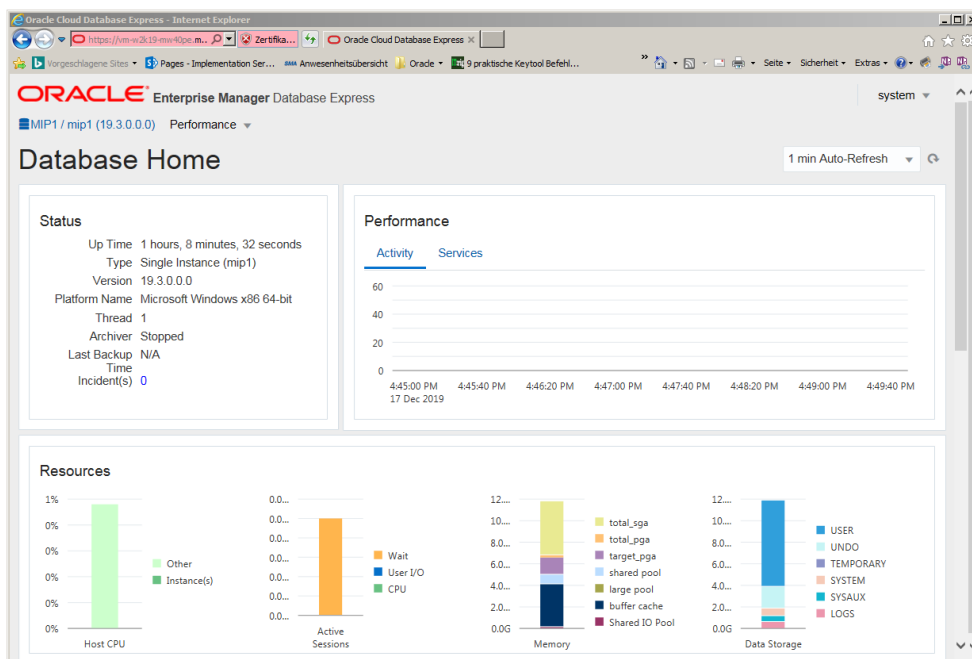
After a successful installation you can test the connection to the Oracle Enterprise Manager (Database Express) by using the Internet Explorer (e.g. to <https://servername:5500/em>).

Note: For using the Oracle Enterprise Manager (Database Express) a web browser with Adobe Flash Player Plugin is needed.

The Oracle Enterprise Manager (Database Express) is only available if the database is running.



Login with user “system” and the defined password



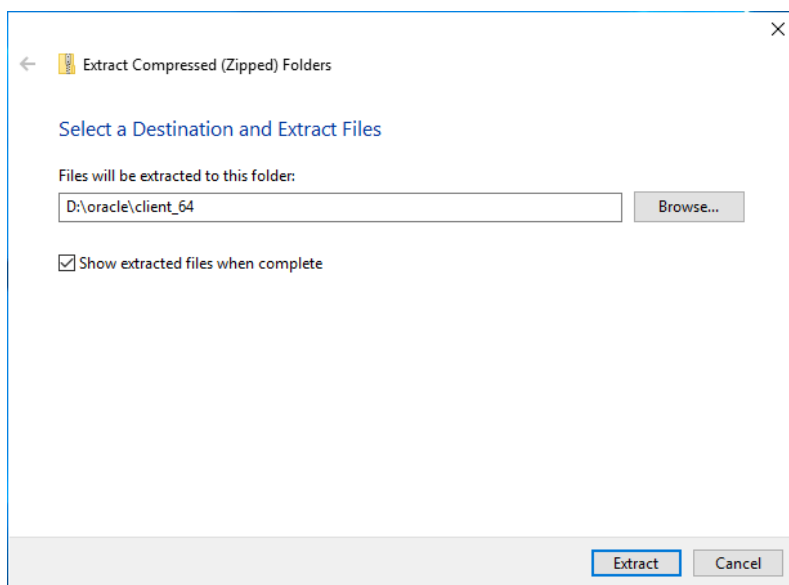
7 Oracle 64Bit Client on Application Server

With **separate application and database servers** for MIP based products the Oracle Client software must **always** be installed on the **application server**.

Login as local user **mipadm**.

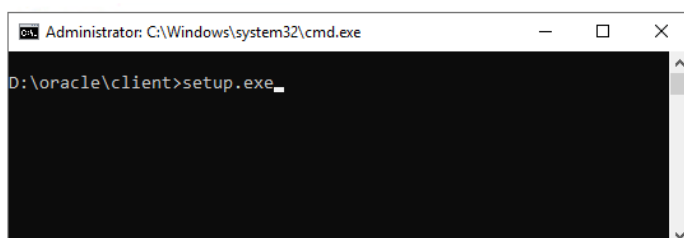
Unblock the downloaded zip file (see chapter [2 Installation Requirements](#)) if the download was done with the Internet Explorer.

Unzip the Oracle Client installation files to a temporary destination directory (e.g.: D:\oracle\client):



Start the installation:

```
D:\oracle\client>setup.exe
```



Oracle Database Client 19c Installer - Step 1 of 6

Specify Oracle Home User

19^c ORACLE Database

Use Windows Built-in Account or specify a standard Windows User Account (not an Administrator account) to install and configure the Oracle Home. This account is used for running the Windows Services for the Oracle Home. Do not log in using this account to perform administrative tasks.

☐ Use Existing Windows User

User Name:

Password:

☒ Create New Windows User

User Name:

Password:

Confirm Password:

The newly created user is denied Windows logon privileges.

☐ Use Windows Built-in Account

Help < Back Next > Install Cancel

User: **orahome**

Password: **ORAhome!!2019**

If you have an existing user account you must use the relevant credentials

Oracle Database Client 19c Installer - Step 2 of 6

Specify Installation Location

19^c ORACLE Database

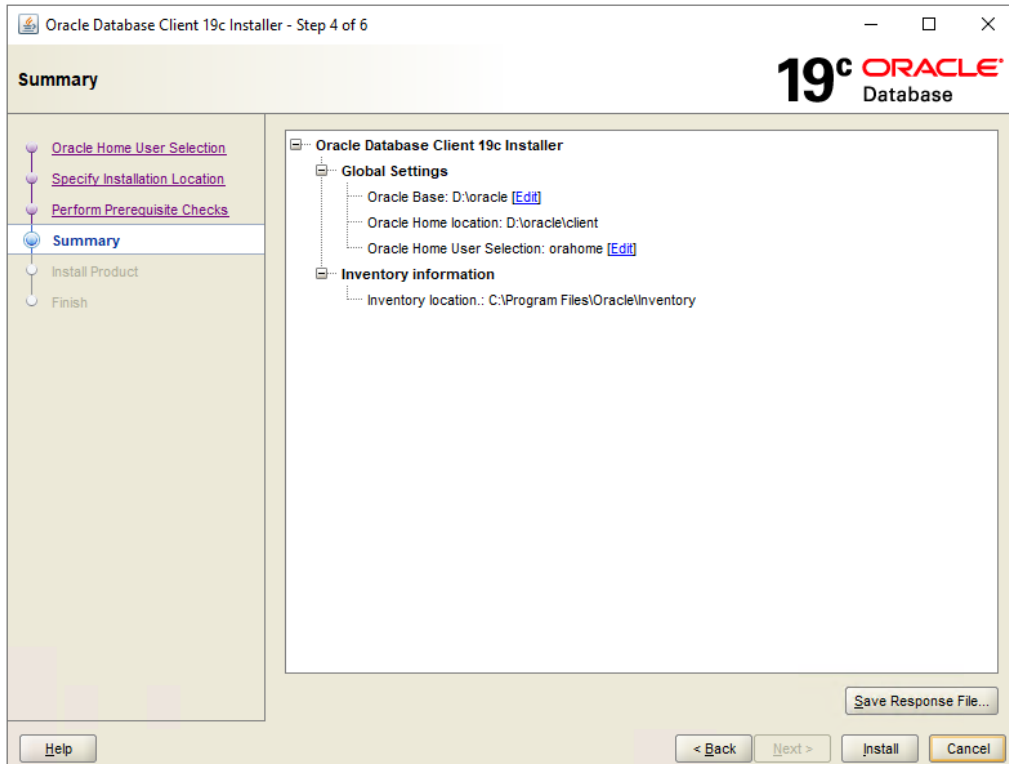
Specify an Oracle base path to place all Oracle software and configuration-related files. This location is the Oracle base directory.

Oracle base: Browse...

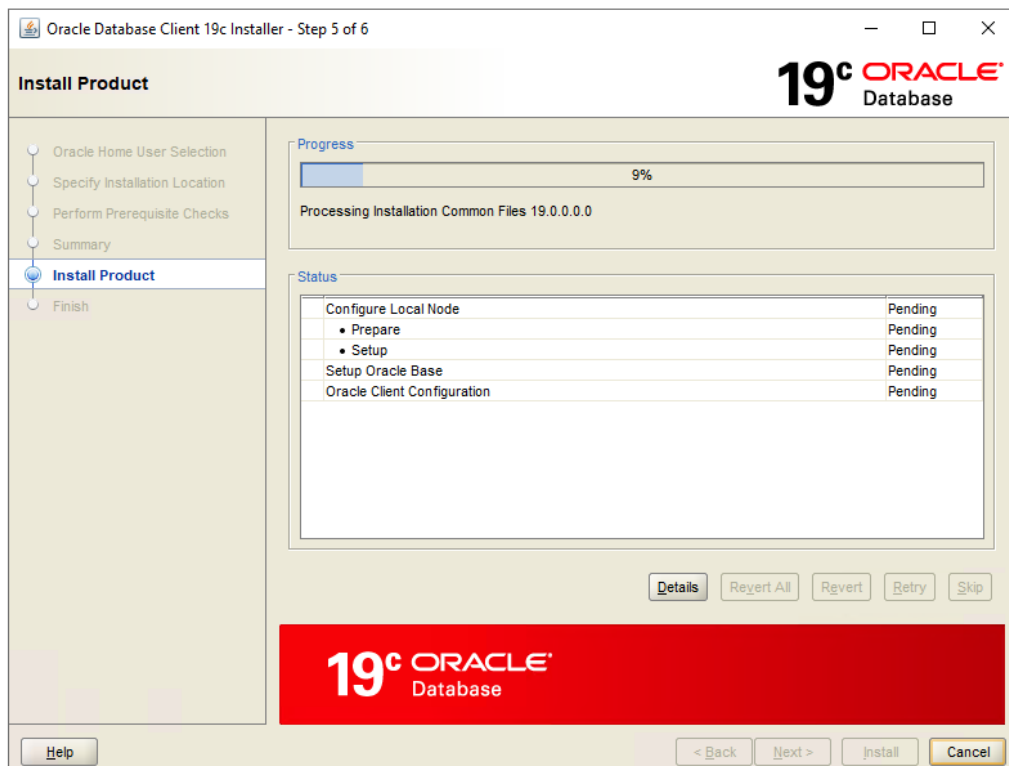
This software directory is the Oracle Database Client home directory.

Software location: D:\oracle\client

Help < Back Next > Install Cancel



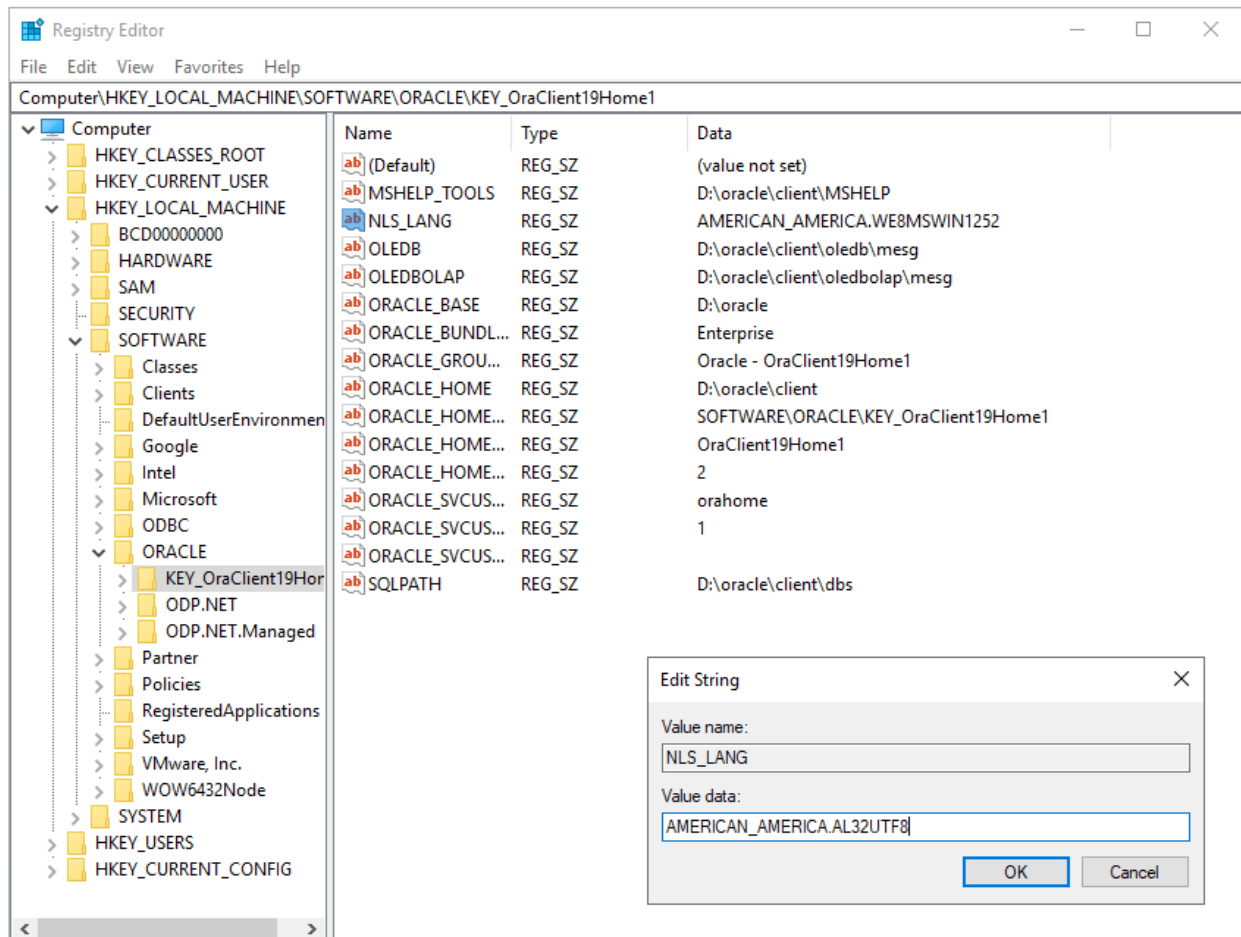
Start the installation





8 Post Installation Configuration

8.1 Registry Settings



HKEY_LOCAL_MACHINE\SOFTWARE\ORACLE\KEY_OraClient19Home1 the contents for key **NLS_LANG** must be changed:

new: **AMERICAN_AMERICA.AL32UTF8**

8.2 tnsnames.ora

Copy the content of d:\mip1\db_sql\oracle19\windows\network\client* to D:\oracle\client\network\admin*

On the HYDRA application server check the **tnsnames.ora** file of the client and add necessary alias configurations for the HYDRA databases (Default: MIP1):

d:\oracle\network\admin\tnsnames.ora

```
exit# tnsnames.ora Network Configuration File: D:\oracle\client\NETWORK\ADMIN\tnsnames.ora
# Generated by Oracle configuration tools.

MIP1 =
  (DESCRIPTION =
    (ADDRESS_LIST =
      (ADDRESS = (PROTOCOL = TCP) (HOST = [IP ADDRESS or FQDN NAME]) (PORT = 1521))
    )
    (CONNECT_DATA =
      (SERVICE_NAME = mip1)
    )
  )

MIP2 =
  (DESCRIPTION =
    (ADDRESS_LIST =
      (ADDRESS = (PROTOCOL = TCP) (HOST = [IP ADDRESS or FQDN NAME]) (PORT = 1522))
    )
    (CONNECT_DATA =
      (SERVICE_NAME = mip2)
    )
  )
```

9 Additional Documents

Oracle Database Installation Guide, 19c for Microsoft Windows:

Database Installation Guide for Microsoft Windows

<https://docs.oracle.com/en/database/oracle/oracle-database/19/ntdbi/index.html>

Database Client Installation Guide for Microsoft Windows

<https://docs.oracle.com/en/database/oracle/oracle-database/19/ntcli/index.html>

Database Administrator's Guide

<https://docs.oracle.com/en/database/oracle/oracle-database/19/admin/index.html>